

Big Data Analytics Pitfalls, Overfitting, and Overparameterization

A. Big Data Analytics Pitfalls: Google Flu Trends

Google Flu Trends (GFT) is a useful example for understanding both the benefits and problems of big data analytics. GFT was developed to estimate influenza activity by analyzing Google search queries. Ginsberg et al. (2009) used CDC influenza-like illness (ILI) data as the target or reference data for their analysis. The CDC surveillance system provided weekly information on the percentage of patients visiting doctors with ILI in nine U.S. regions. The researchers then compared these weekly CDC ILI levels with millions of weekly Google search queries to see which searches followed a similar pattern. They tested about 50 million search queries, measuring how strongly each query was correlated with the 1,152 CDC ILI data.

Google Flu Trends had several important benefits:

First, it used Google search data to estimate influenza activity quickly, without waiting for traditional surveillance reports. Traditional CDC surveillance could have a reporting delay of one to two weeks, while Google search data were available much more rapidly. This meant that GFT could potentially provide an early warning of increases in influenza activity.

Second, Google search data provided a very large amount of information from people's online activities. By comparing millions of search queries with CDC influenza data, the researchers identified search patterns that were strongly related to ILI. The final model showed a strong relationship with CDC data and could estimate influenza activity one to two weeks ahead of the traditional reports.

However, GFT also shows several *problems* with big data analytics Lazer et al. (2014)

Big data hubris means assuming that having a very large amount of data will automatically produce better results. GFT shows why this can be a problem. A search for “flu symptoms” does not necessarily mean that the person actually has the flu. They might be searching because they are curious, worried about someone else, or responding to news about an outbreak. Therefore, a very large dataset does not automatically mean that the data accurately measure the phenomenon we are interested in.

Lazer and his colleagues argued that big data should supplement, not replace traditional data and scientific methods. The Google Flu case showed that researchers must examine how the data are generated, validate the results, and check whether the relationship between the data and the real-world phenomenon remains stable over time. Thus, having more data does not guarantee better or more accurate results.

Next issue is the changing **algorithm dynamics**. Google's search algorithm and users' search behavior were changing, which changed the relationship between search queries and flu activity. Google Flu Trends assumed that the same search patterns would always represent the same level of flu activity. When the search patterns changed, the model became less accurate and sometimes predicted much more flu than was actually reported by the CDC.

GFT also shows why **transparency and replicability** are important. Google's search data and algorithms were private and could change over time. This made it hard for other researchers to reproduce the model and understand why its predictions became less accurate.

The Google Flu case shows that big data can be useful, but having a large amount of data does not always mean the results are accurate. Lazer and his colleagues highlighted these problems, where researchers may rely too much on large datasets and ignore changes in algorithms and people's behavior. So, big data should support, not replace, traditional data and scientific methods. Scientific models should provide enough information for others to check the data, methods, assumptions, and results.

B. Overfitting and Overparameterization: Google Flu Trends

Overfitting means that the model memorizes the old data instead of learning the general pattern. Because of this, the model may work very well on the data it was trained on but perform poorly when it gets new data.

For example, imagine a student memorizes the answers to practice questions instead of understanding the topic. They may get all the practice questions correct, but struggle with new questions. Similarly, Google Flu Trends learned patterns from past Google searches that matched flu activity, but some of those patterns were only accidental relationships. When new search patterns appeared, the model could not predict actual flu activity accurately.

Overparameterization means a model uses too many variables or factors to make a prediction, making the model unnecessarily complicated. Instead of using a few meaningful variables such as “flu symptoms,” “fever,” and “flu medicine,” the model uses thousands of different Google search terms. Some of these terms may have little or no real connection to flu, but the model still uses them to make predictions. Using too many variables in a model can make the model unnecessarily complicated and thus, more likely to find accidental relationships.

An example R chart that shows overfitting:



It is a built-in R air quality dataset containing daily air-quality measurements in New York. For this chart, temperature (Temp) is used to predict ozone level (Ozone). Rows with missing values are removed, and the remaining data are divided into training data (70%) and test data (30%). Polynomial models from degree 1 to degree 10 are then fitted using the training data.

The chart shows **training error and test error** as the **polynomial degree increases**. As the degree increases, the *training error usually decreases* because the model becomes more flexible and fits the training data better. However, after a certain point, the *test error begins to increase*. This means the model has started learning the specific patterns and noise in the training data instead of the general relationship between temperature and ozone.

In the chart, as the polynomial degree increases, the model gets more flexible and can fit the training data more closely. But if the degree becomes too high, the model may start fitting random noise in the data, this causes overfitting.

References

Ginsberg, J., Mohebbi, M. H., Patel, R. S., Brammer, L., Smolinski, M. S., & Brilliant, L. (2009). *Detecting influenza epidemics using search engine query data*. **Nature**, **457**, 1012–1014. <https://doi.org/10.1038/nature07634>.

Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). *The parable of Google Flu: Traps in big data analysis*. **Science**, **343**(6176), 1203–1205. <https://doi.org/10.1126/science.1248506>.